

Bucket Wheel Excavators

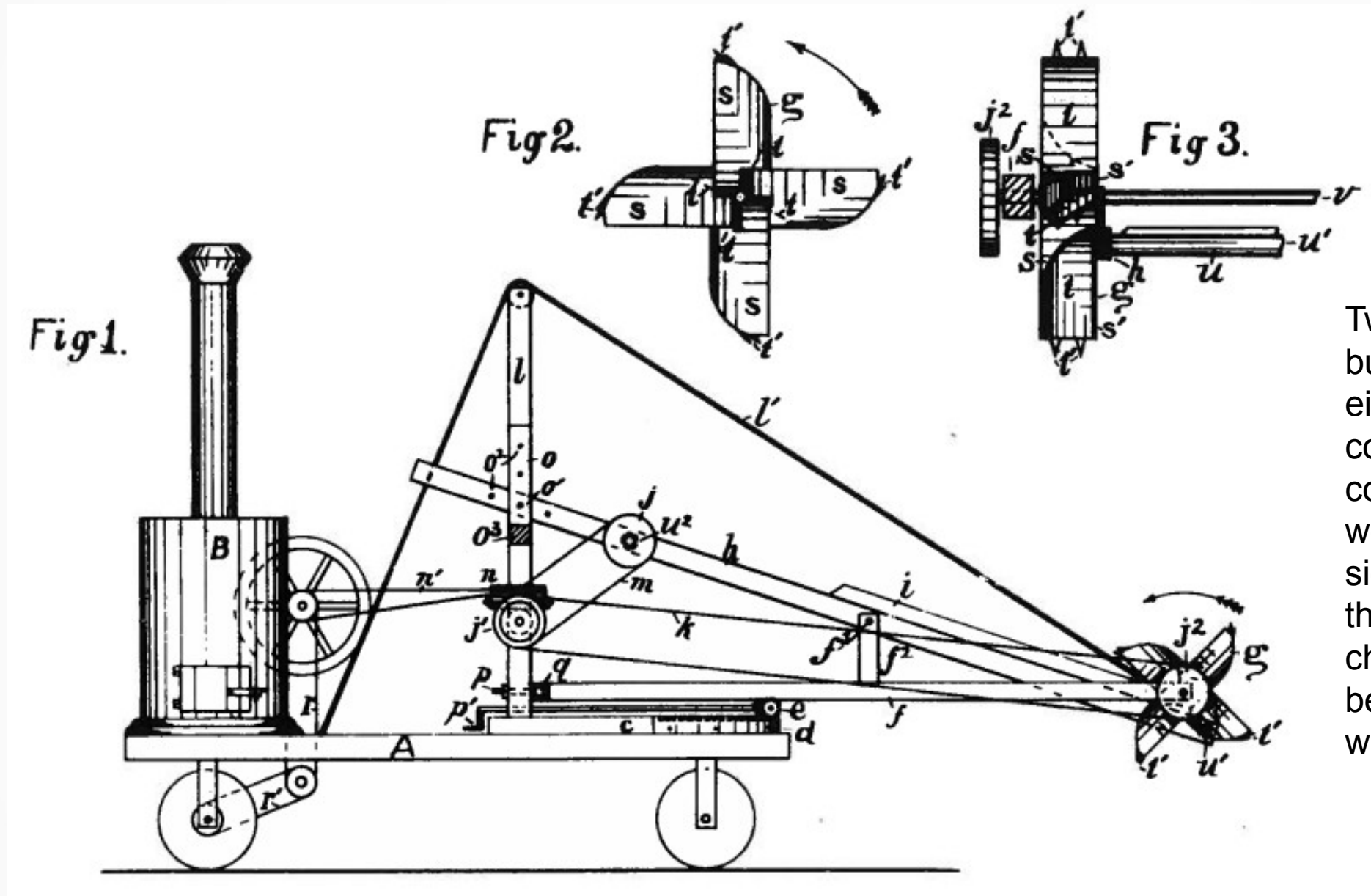
(Continuous Mining System)

Motivation of Bucket wheel Excavators



The Norias of Hama are a number of norias ("wheels of pots") along the Orontes River in the city of Hama, Syria. The earliest evidence for norias in Hama suggests they were developed during the Byzantine era, although none of the norias in Hama today predate the Ayyubid period (12th/13th centuries). The surviving norias date back to the medieval Islamic period, when norias as large as 20 meters in diameter were built, with 120 water collection compartments and capable of raising more than 95 litres of water per minute.

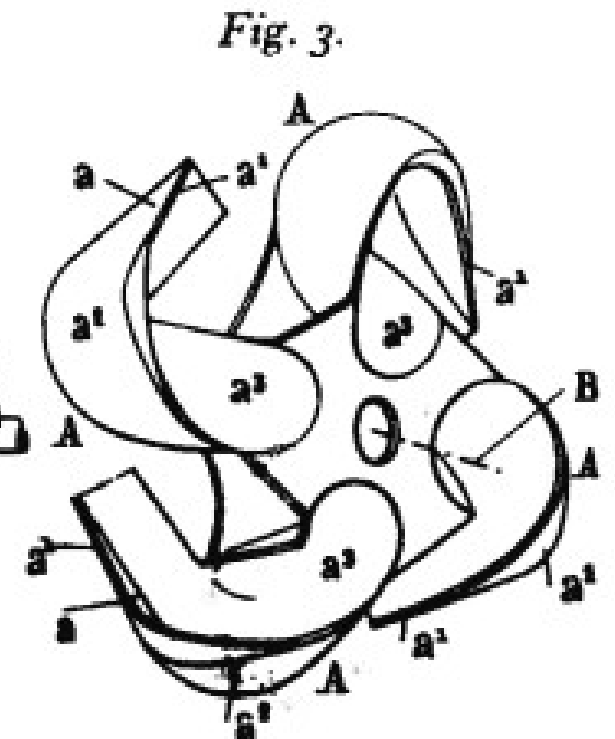
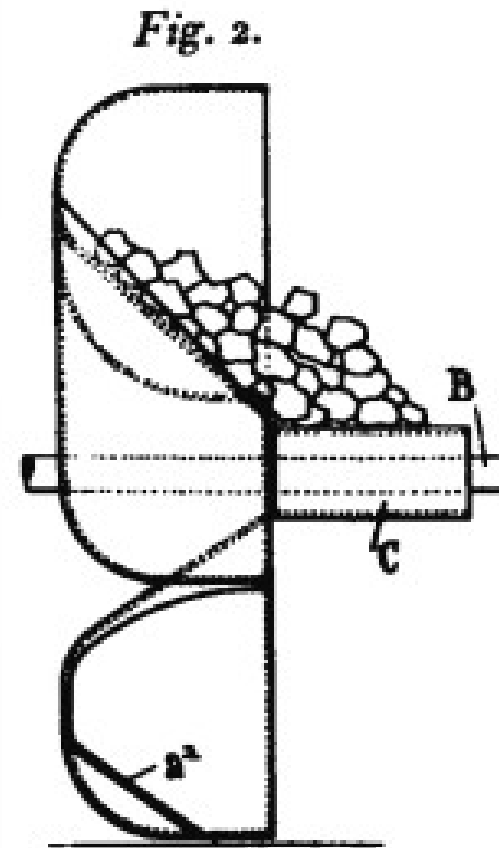
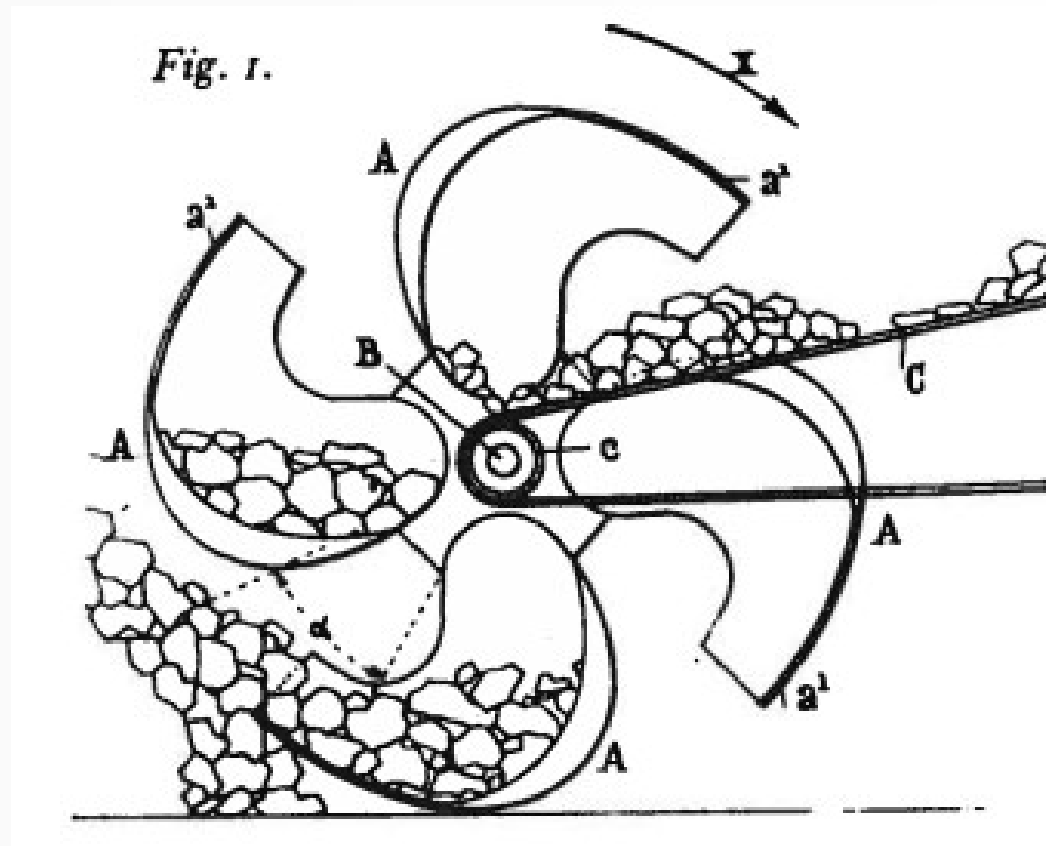
Early design of Bucket Wheel Excavators



Two free cutting bucket wheels on either side of a continuous conveyor. These were emptied sideways through separate chutes onto the belt near the wheel hub.

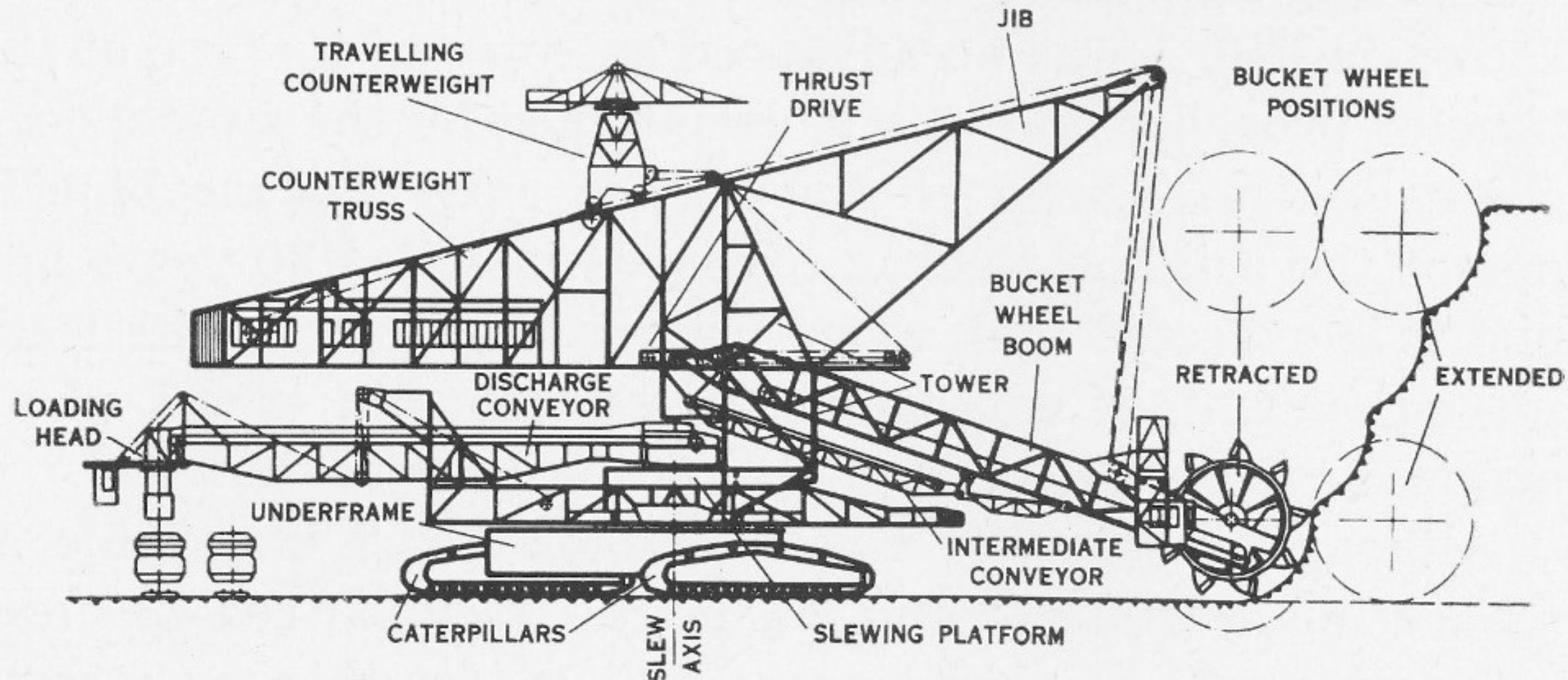
Developments in Bucket Designs

The buckets were arranged on a common axis of rotation in such a way that, while turning they dumped the material automatically on to a conveyor belt.



German Patent DPR no. 272,118, Mar 2, 1913 : George Eugene Auguste Schwander, Mont-beliard

(Rodgers, 1960, p.138)

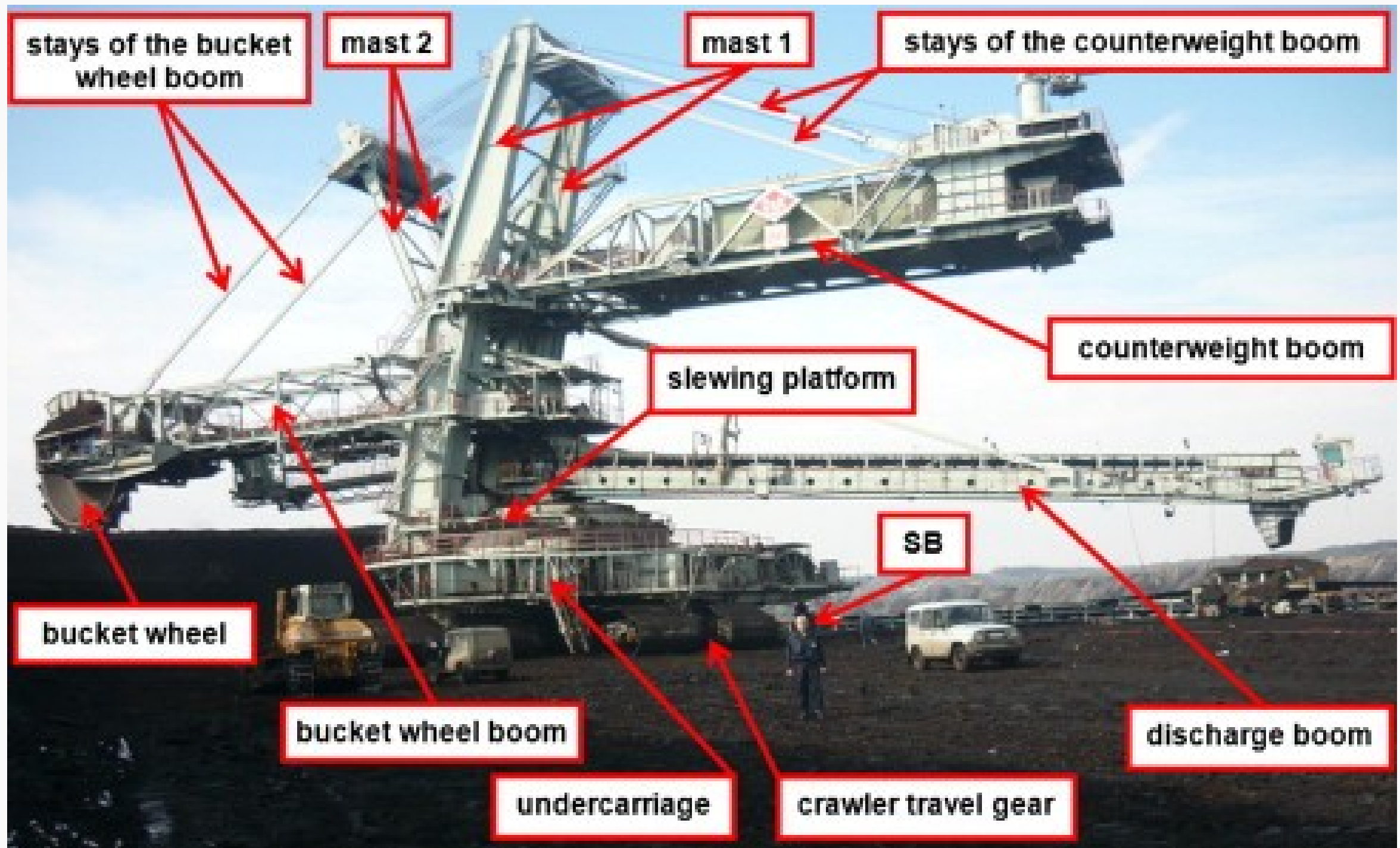


Manufacturer
TAKRAF
generally
refers to it as
an excavator
of the type
SRs 8000

- Largest and Heaviest land vehicle (GB world Record)
- Weight : 14,196 tonne (31.3 million-lb)
- Top speed : ~ 0.33 mph (0.53 km/h)
- 18 buckets, each with a volume of 6,600 litres (1,743 US gal)
- capable of shifting 240,000 m³ (8.475 million cu ft) of earth in a single day- This is the equivalent of around 3 million bathtubs' worth of soil.

Bagger 293

Richard Bradbury



Classification of BWE

Type of bucket

- Cell type
- Cell-less type
- Semi cell type
-

Propel Mechanism

- Crawler mounted
- Rail mounted

Type of cutting boom

- Crowding type
- Non-crowding type

Type of discharge boom

- Telescopic type
- Non-telescopic
- Direct casting type

Type of cutting/digging

- High digging type
- High and deep digging type

Cell Type Bucket Wheel

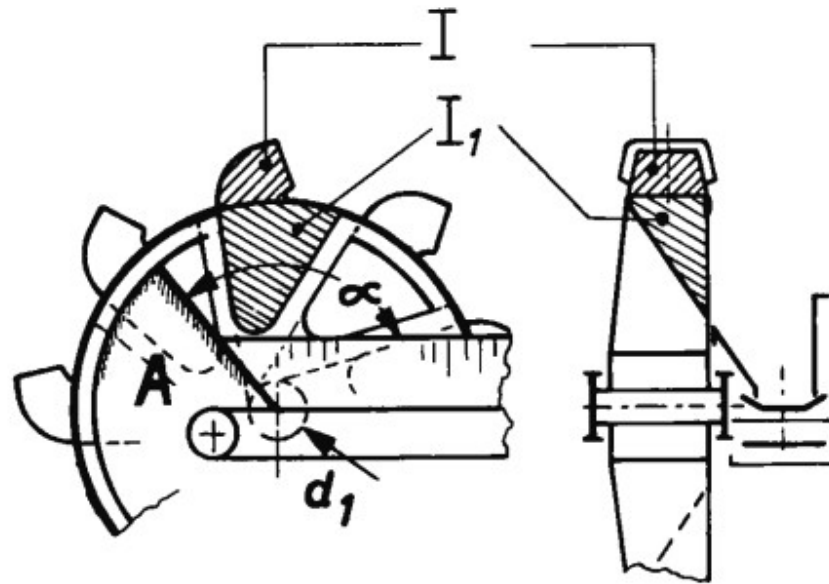


Fig. 6-1 (a): Cell-type bucket wheel.
 I = capacity of bucket; I_1 = capacity of discharge cell;
 A = cover plate; d = diameter of imaginary cylinder formed by the inclination of the discharge chutes; α = angle of discharge opening.



Cell Less Bucket Wheel

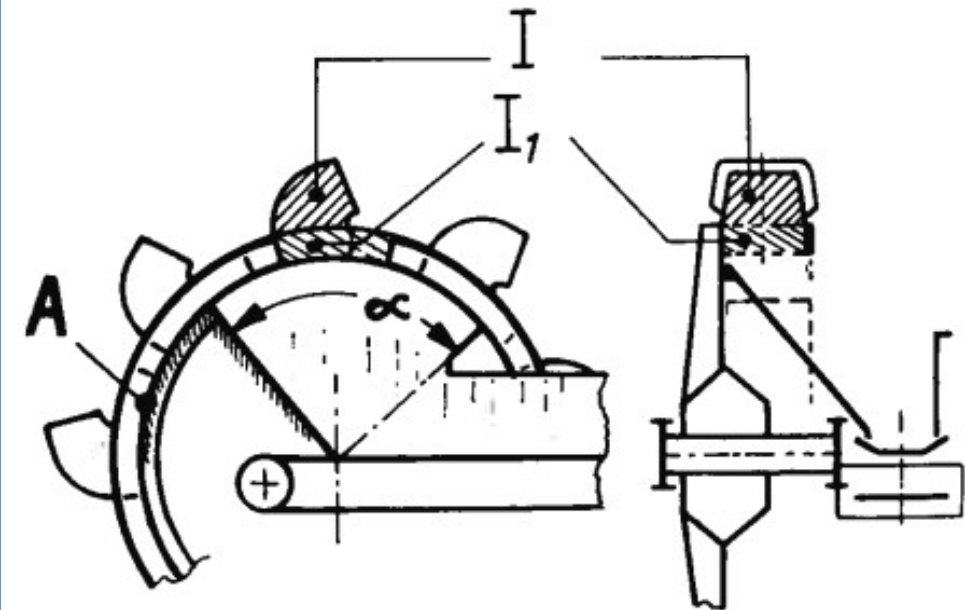
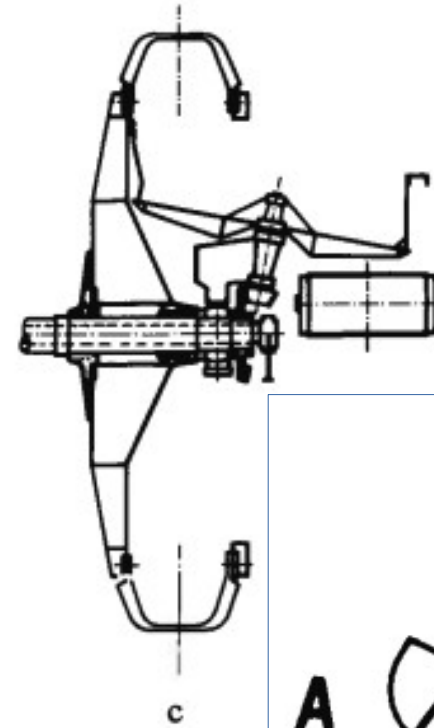
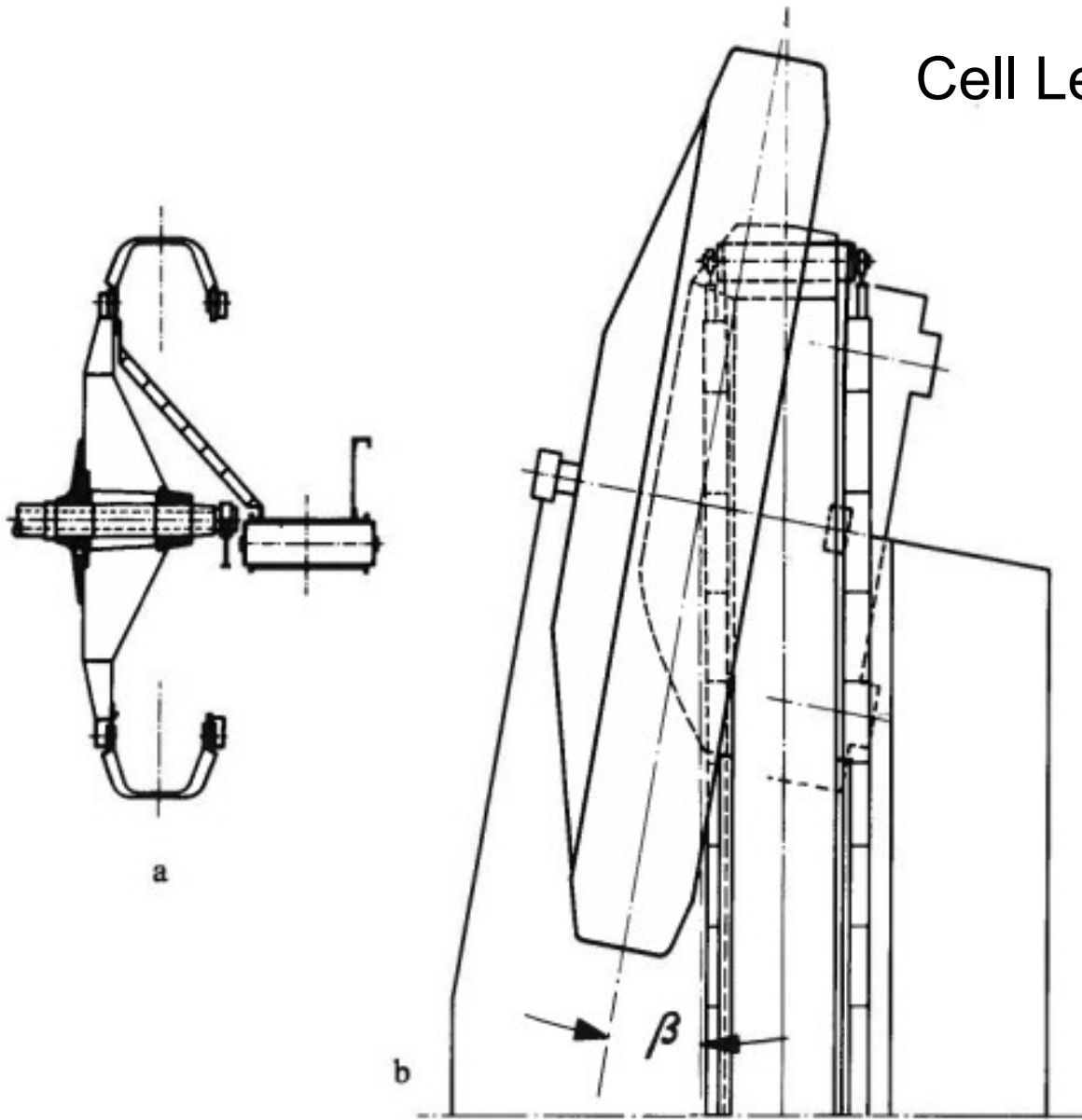


Fig. 6-3: Cell-less bucket wheel discharge arrangements.
(a) slope chute; (b) horizontal tilt (β) of the wheel axis; (c) rotary disc feeder.

Fig. 6-1 (b): Cell-less bucket wheel.
 I = capacity of bucket; I_1 = capacity of ring chute; A = ring space cover plate.



Fig. 6-5: Cell-less wheel with roller discharge.

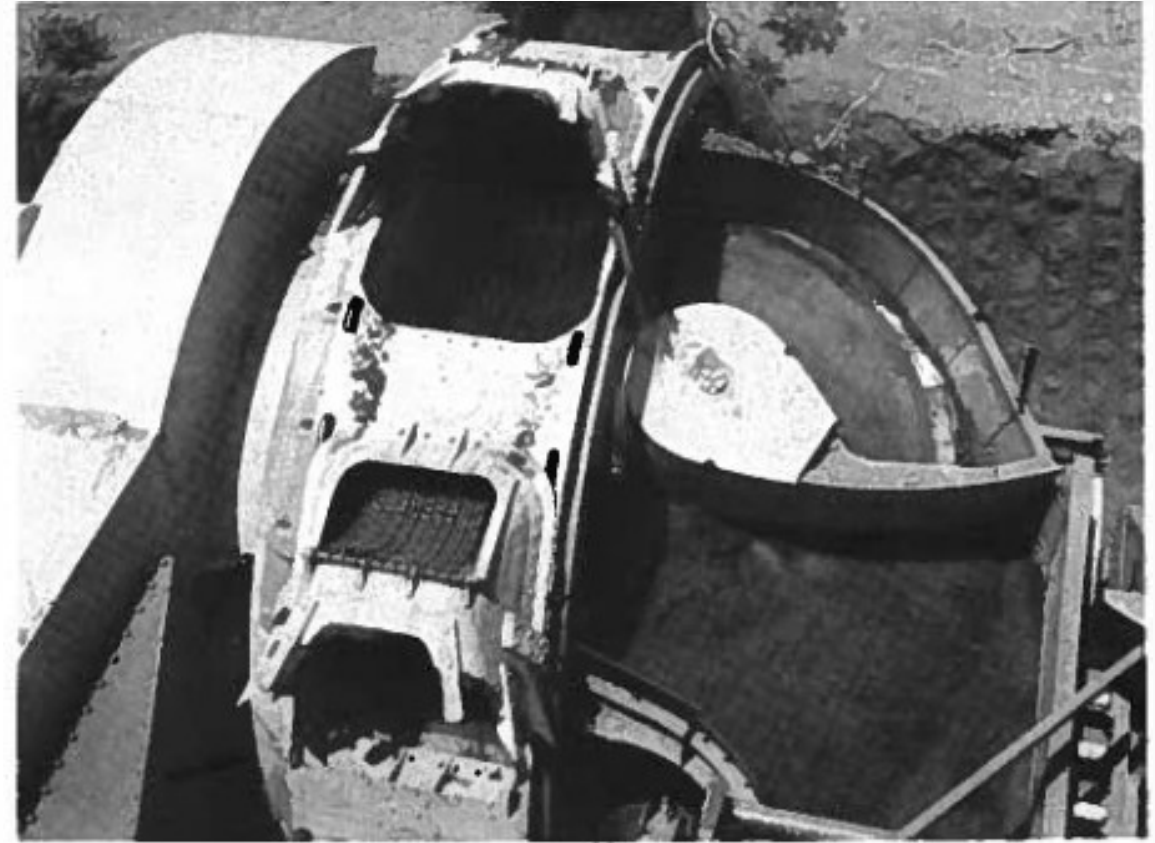


Fig. 6-6: Rotary disc feeder with adjustable plough for high and deep cut (shown is emptying process during high cut) (O & K/LMG).

Semi Cell Type Wheel

Introduced for wheel dia in excess of 17 m.
Compared to the other two types it has the advantage of greatest bucket fill

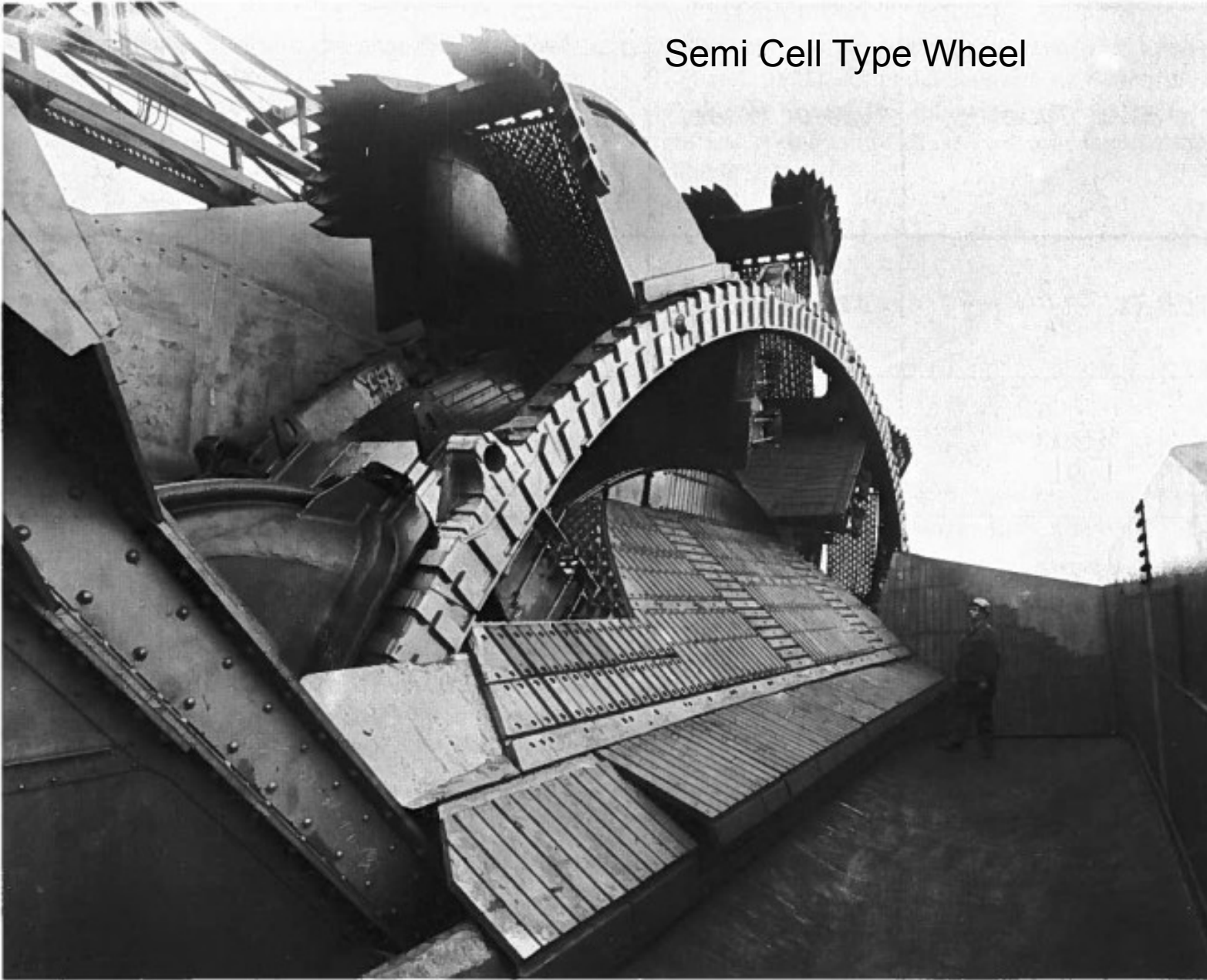
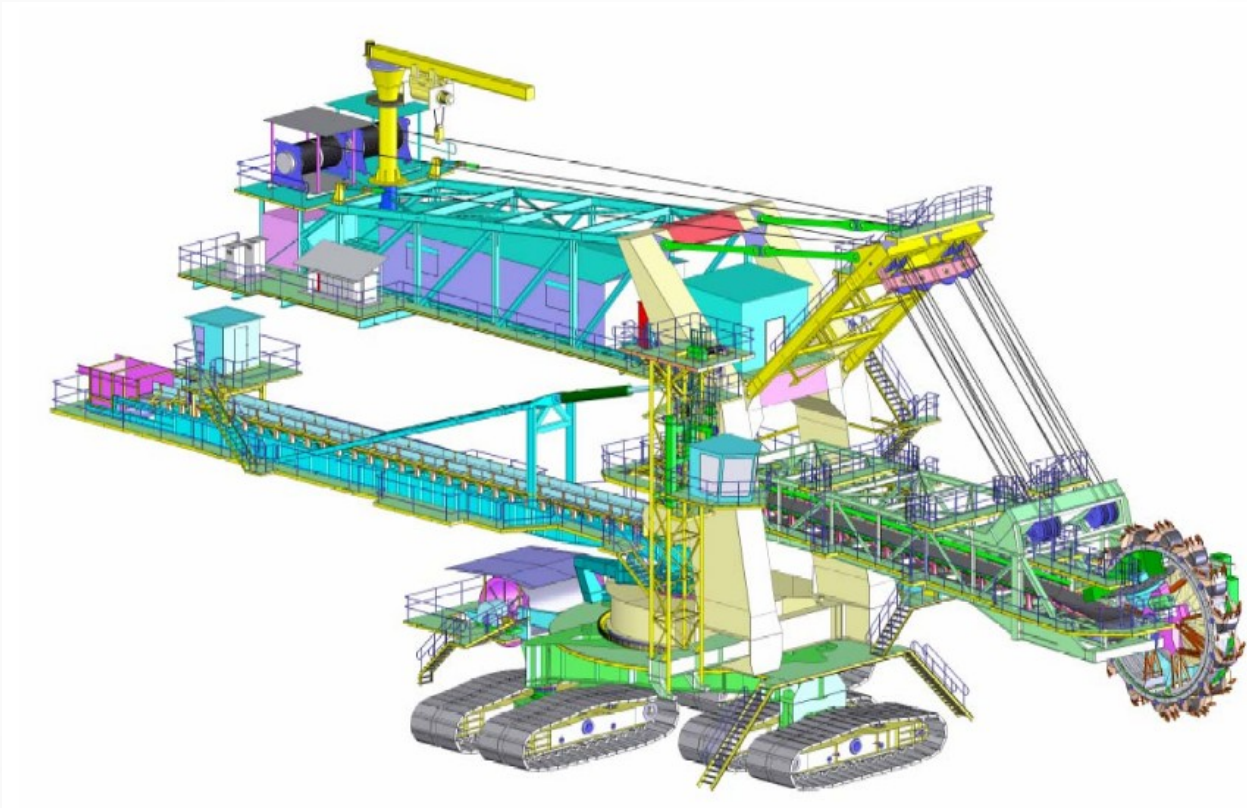


Fig. 6-8: Semi-cell type wheel of BWE SchRs $\frac{4500}{14}$ – 41 (O & K/LMG).

Bucket Wheel Excavator Type: VABE 1400-NLC-II



VOEST-ALPINE Materials Handling

Bucket capacity	1400 l
Bucket wheel diameter	10.5 m
Number of buckets	14
Discharges	70 min -1
Theoretical capacity	6500 lm3/h
Materials to be handled	Overburden
Slewing range of superstructure to discharge boom	+/- 105 degree
Slewing range of superstructure to underframe	+/- 105 degree
Belt width	2.0 m
Belt speed	4.7 m/s
Ground pressure	120 kPa
Min. turning radius	50 m
Travelling speed	8 m/min
Maximum gradient	in operation during relocation
	1 : 20 1 : 10
Power of bucket wheel motor	2 x 750 kW
Total installed power	3800 kW
Year of delivery	2007 (3 units)
Location	Neyveli, India

Application

Bucket wheel excavator finds it application when:

Applicability of BWE

- 1)The formations are soft to medium hard.
- 2)The formations are free of geological disturbances.
- 3)The formations are free of boulders.
- 4)The formations are such that they not clog the wheel, transfer point, belts etc because of stickiness or pressure of buried vegetations.
- 5)Selective mining of inter bedded ore and waste is necessary.
- 6)Continuous working is possible.
- 7)Reserve is large and amenable to high production rates.
- 8)Deposit should be flat, which can be worked in simultaneously backfilling.

Disadvantages of BWE

- 1.Requirement of high capital investment as the machine is costly.
- 2.Require a very high reserve that should be amenable to high rate of extraction over a long mine life.
- 3.The system of extraction is rigid in the sense that it is not amenable to flexibility of operations compared to other conventional mining with shovel and dumper.

Conditions that can rule out the use of BWE's include

- 1) Presence of uniformly hard materials that fragment poorly with zero-throw blasting (limited drilling and blasting).
- 2) A high percentage of large and un-sized boulders that is too big for the buckets to handle during cutting.
- 3) Presence of buried vegetations or such other things that would clog the cutting wheel, transfer point, belts etc.
- 4) Presence of inter-bedded hard dense rock that slows the production unduly and increases maintenance costs.

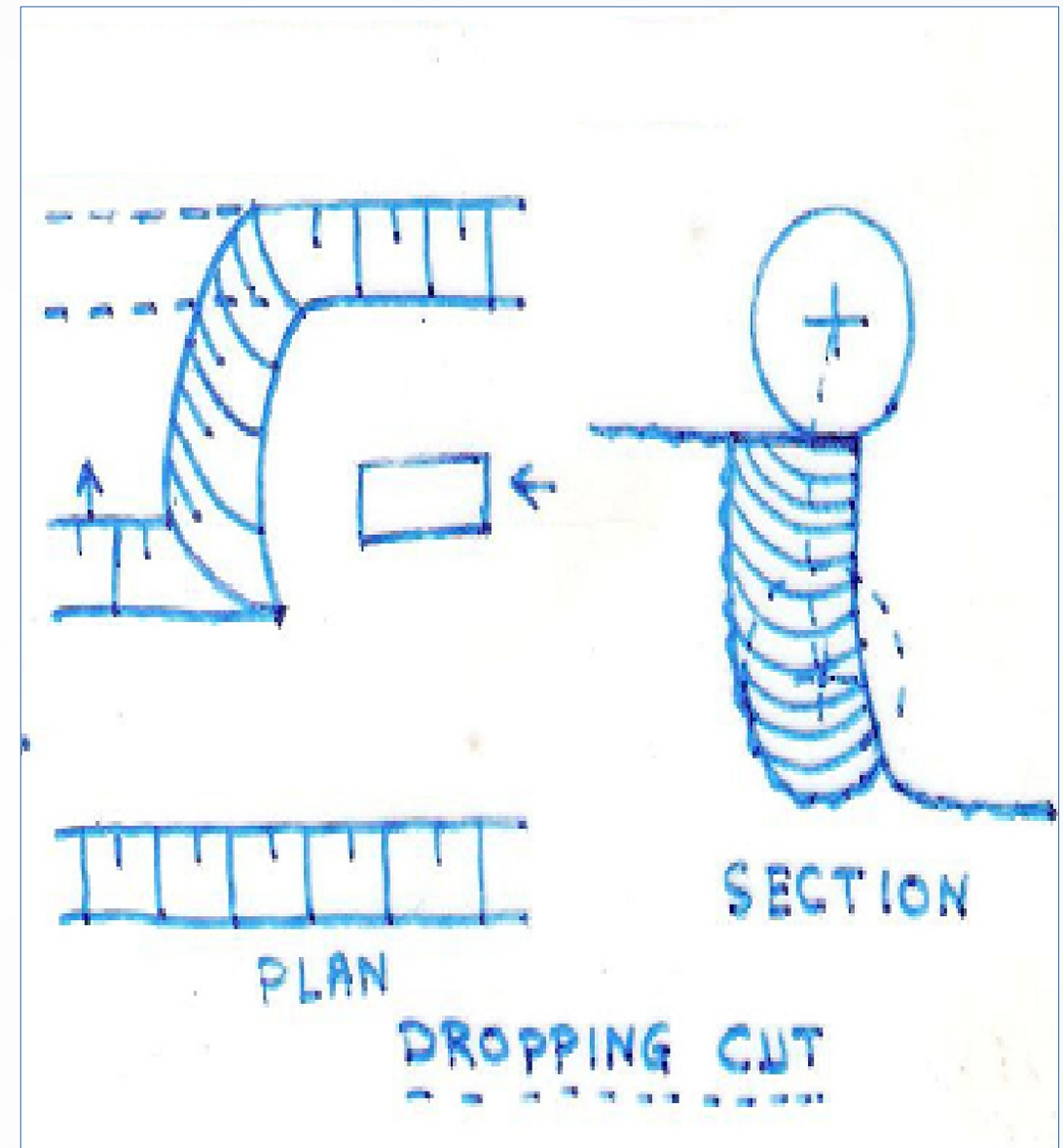
Advantages of BWE

1. More stable pit slopes for both high wall and spoil heap.
2. Availability of wider working spaces.
3. Greater amounts of developed reserves exposed by operating on wide benches.
4. Back filled spoil piles are more even in size and contour than with other excavators.
5. Close control in selective mining of inter bedded ore and waste (in lateral block method).
6. Ability to deliver material on the same, above or below the working level.
7. High and/or deep cuts can be handled depending on machine design.
8. Comparatively smooth and level working benches.
9. Cost of excavation of material per ton or per m³ is generally lowest.
10. Minimum load fluctuation due to uniform operation.
11. Large fleet of mobile transport equipment requiring a commensurate manpower can be totally avoided.

Dropping Cut

In the dropping cut, the final slope of the face is defined by the vertical arc of the cutting wheel about its boom pivot point on the machine main frame. Cuts are made by the bottom arc of the cutting wheel progressing from the top to the bottom of the bench face as the wheel slew from side to side.

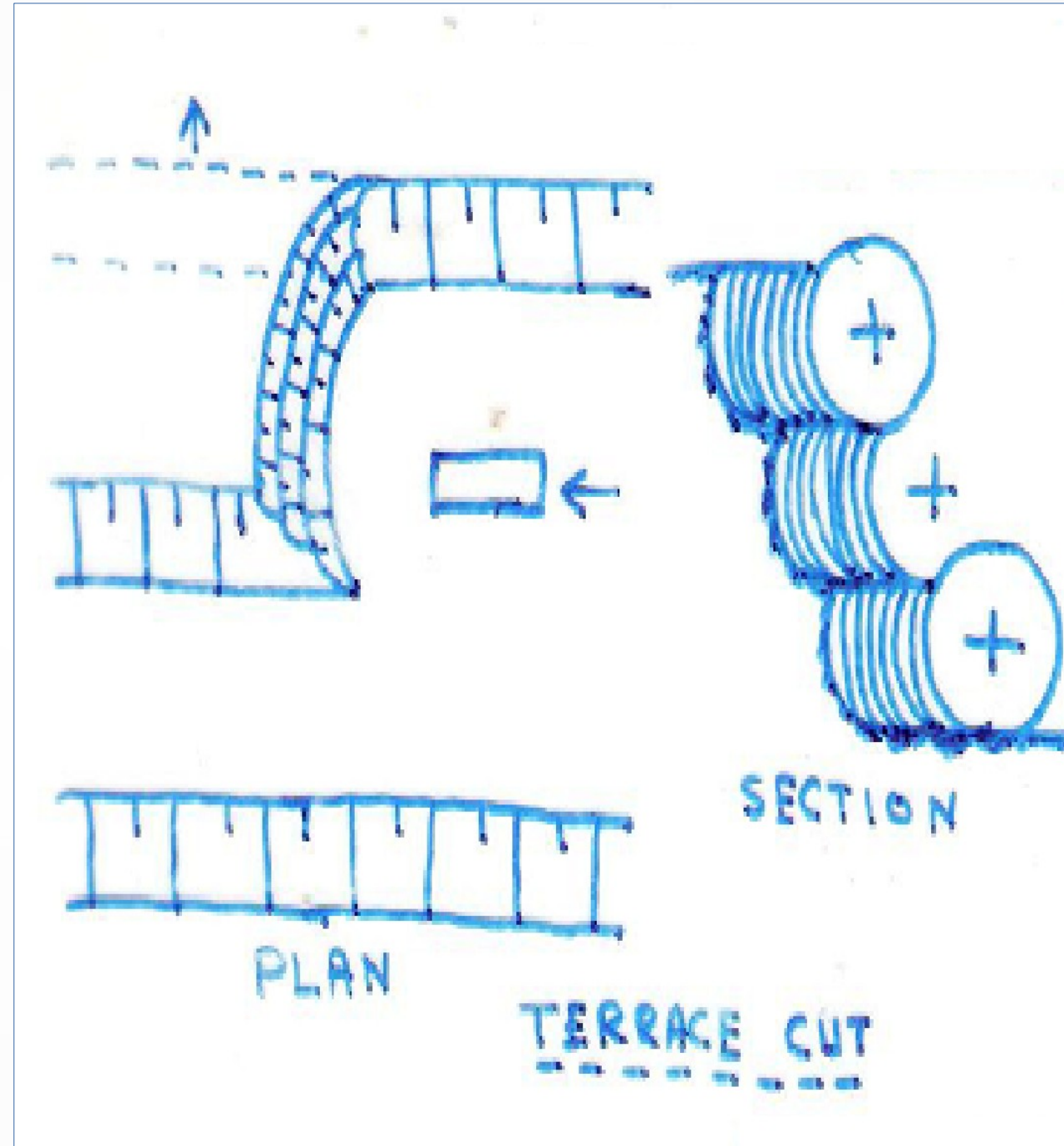
To start the cut sequence the bottom arc of the cutting wheel is placed at one end of face to create the first cut height and then the wheel is slew from side to side. Once a cut is made over the whole face width, the cutting wheel is dropped down for another cut height and the process is repeated till the whole bench height is excavated. After a cut has been made over the entire face, the BWE is repositioned by advancing to the selected cut depth.



Terrace Cut

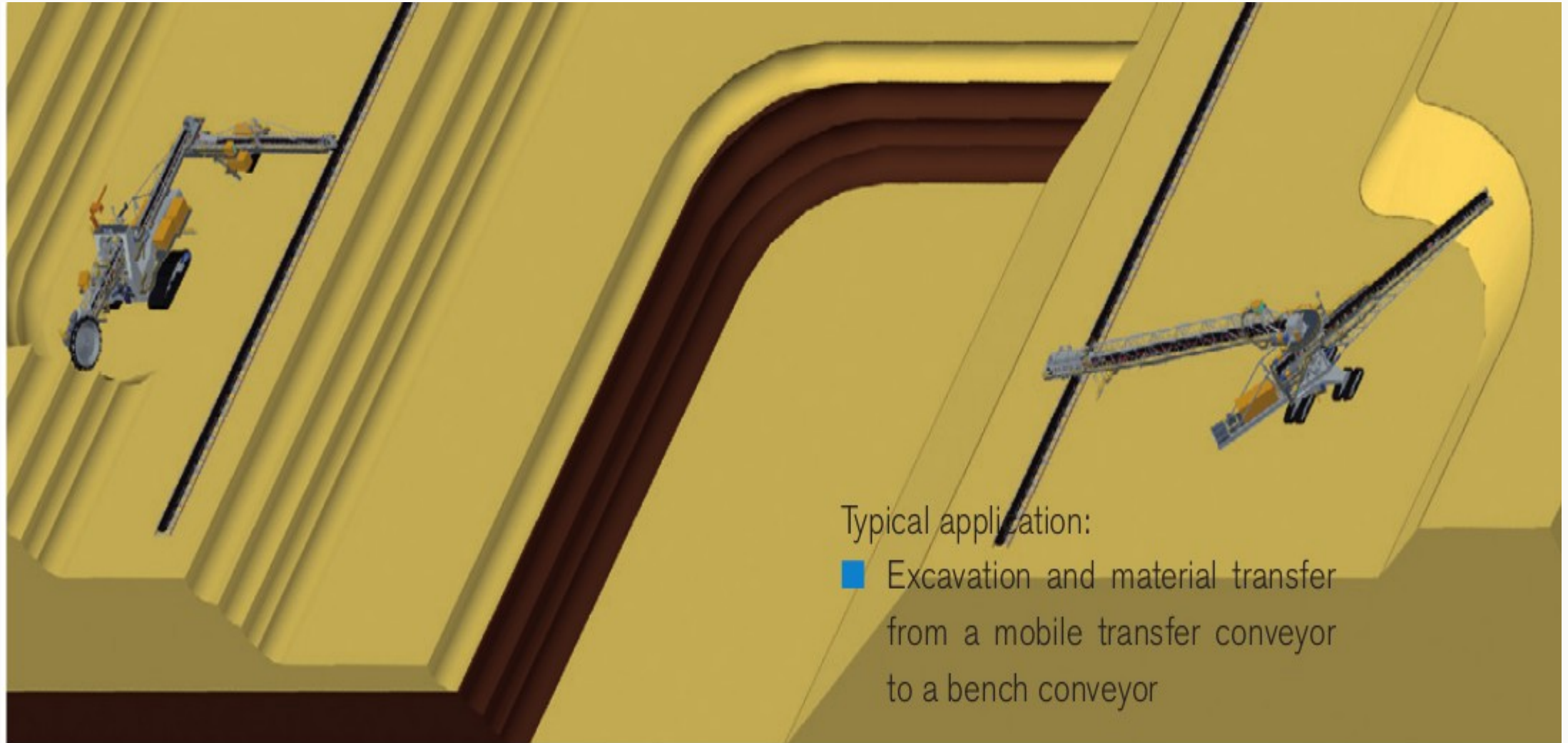
In the terrace cut, the face slope is established by a series of small bench type cuts made by the forward arc of the cutting wheel. To start excavation of a terrace cut, the BWE engages the forward arc of the cutting wheel at one end of the slice to be cut to establish the first cut depth by propelling the machine forward and then the wheel is slewed from one end to the other end of the face (swing pass).

The machine is propelled forward after each swing pass to provide the cut depth and this is repeated until the desired terrace depth is achieved. The machine is then repositioned (by propelling backward) for the start of the next cut sequence. In this way the whole bench height is excavated in required number of terraces. Once the lowermost terrace is excavated the BWE is advanced by a distance equal to the selected terrace depth to start a fresh sequence of cutting terraces.



	Terrace Cut	Dropping Cut
Merits	1)It leaves a comparatively level floor. 2)The face slope can be controlled by maintaining designed space between two successive terraces. 3)For the same BWE it can excavate higher benches compared by dropping cut.	1)It minimizes lumps and improves efficiency in hard materials of a stable slope. 2)It provides easier operator control of the cutting slice depth and overall slope. 3)It requires less machine travel compared to that in terrace cut.
Demerits	1)Terrace cutting in the upper slices tends to produce large lumps when digging into agglomerated, semi-compacted material. 2)It requires more travelling compared to that in dropping cut. 3)The forward and reverse propelling sequence of the BWE may tear up the floor under marginal conditions.	1)It leaves lumpy floor. 2)The face slope cannot be controlled it is defined by the vertical arc of the cutting wheel. 3)It normally requires more power compared to that in terrace cut. 4)For the same BWE it cannot excavate as higher benches as by terrace cut.

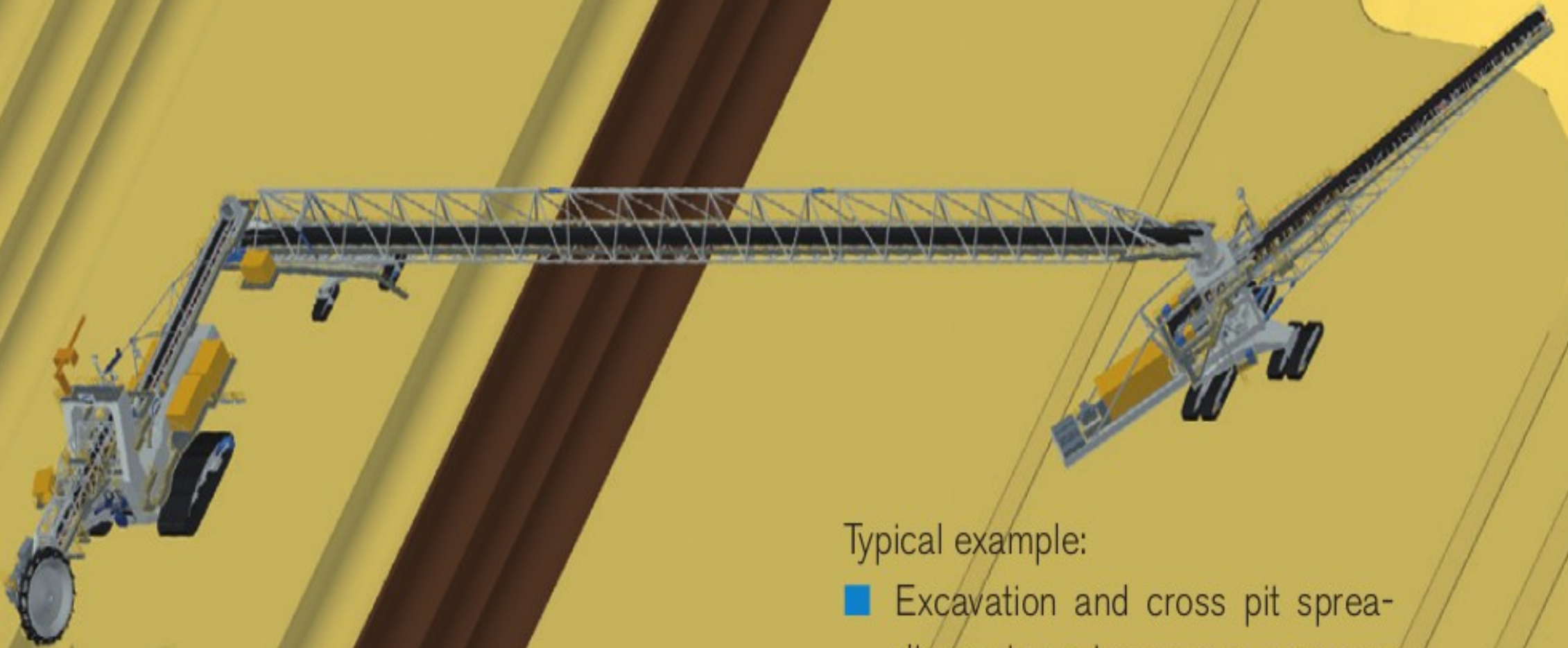
Complimentary equipment and systems



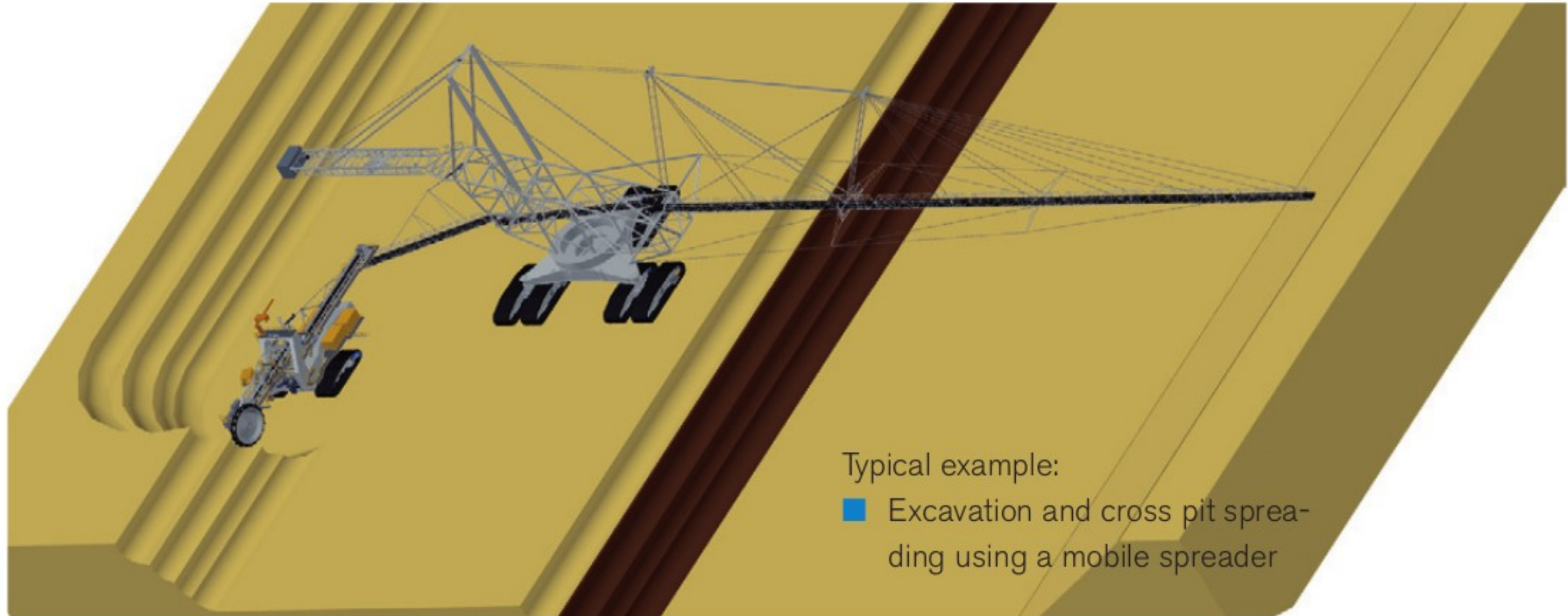
Complimentary equipment and systems

Typical example:

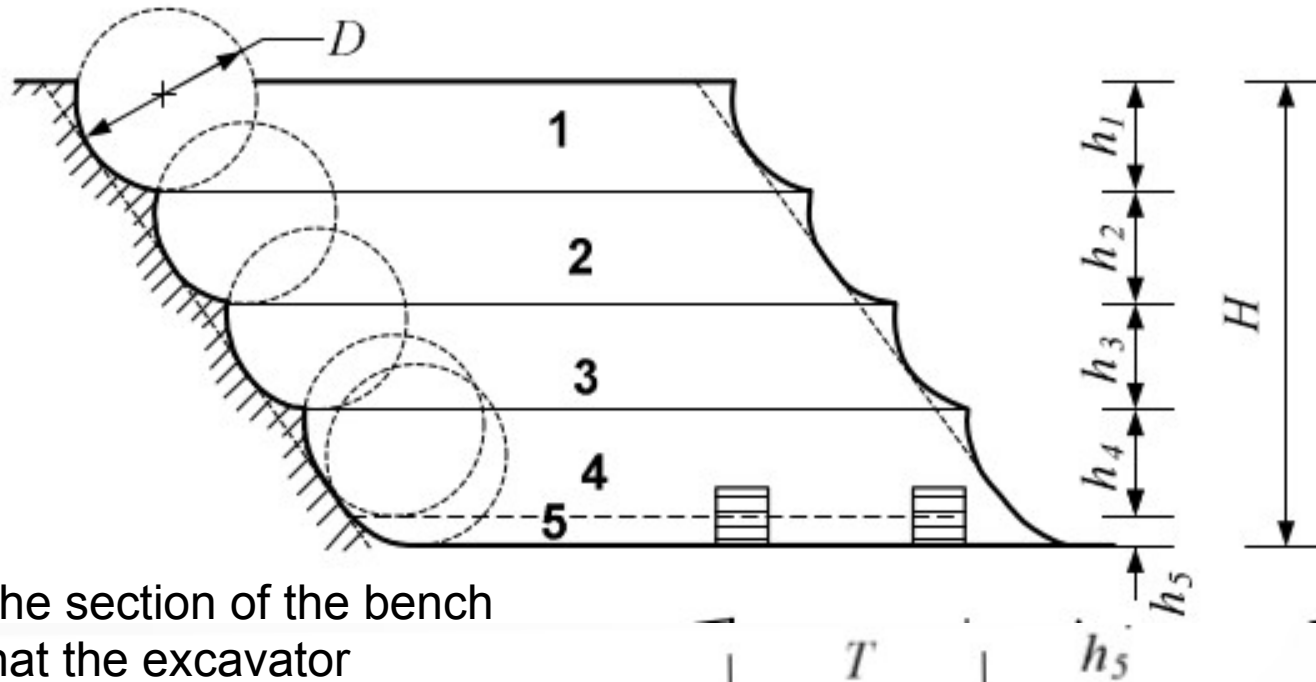
- Excavation and cross pit spreading using a transverse conveyor bridge and connected spreader



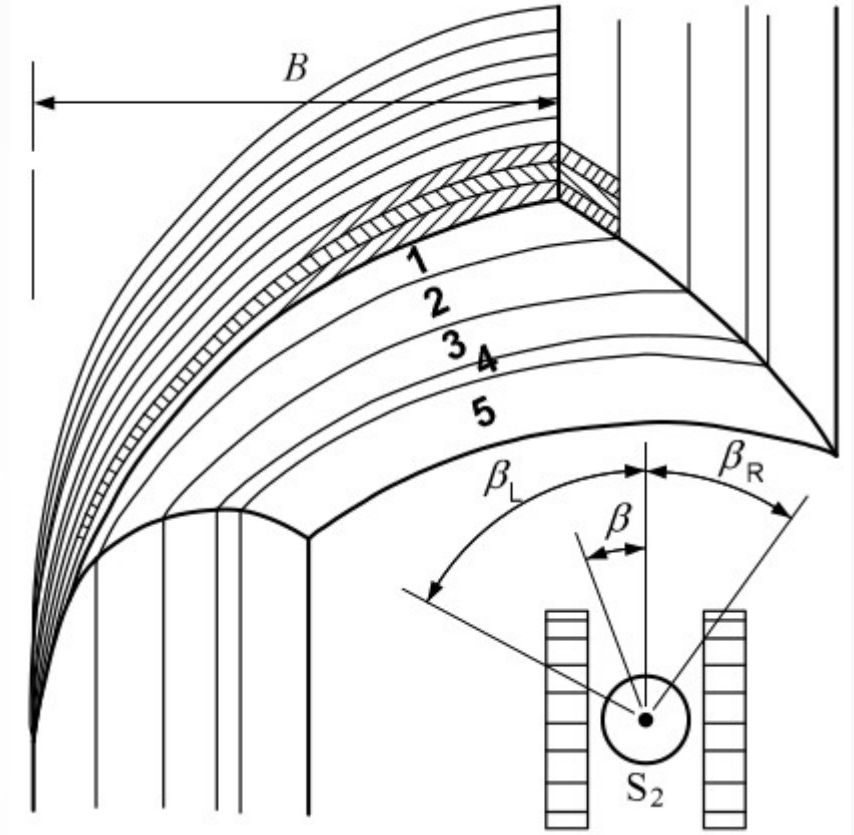
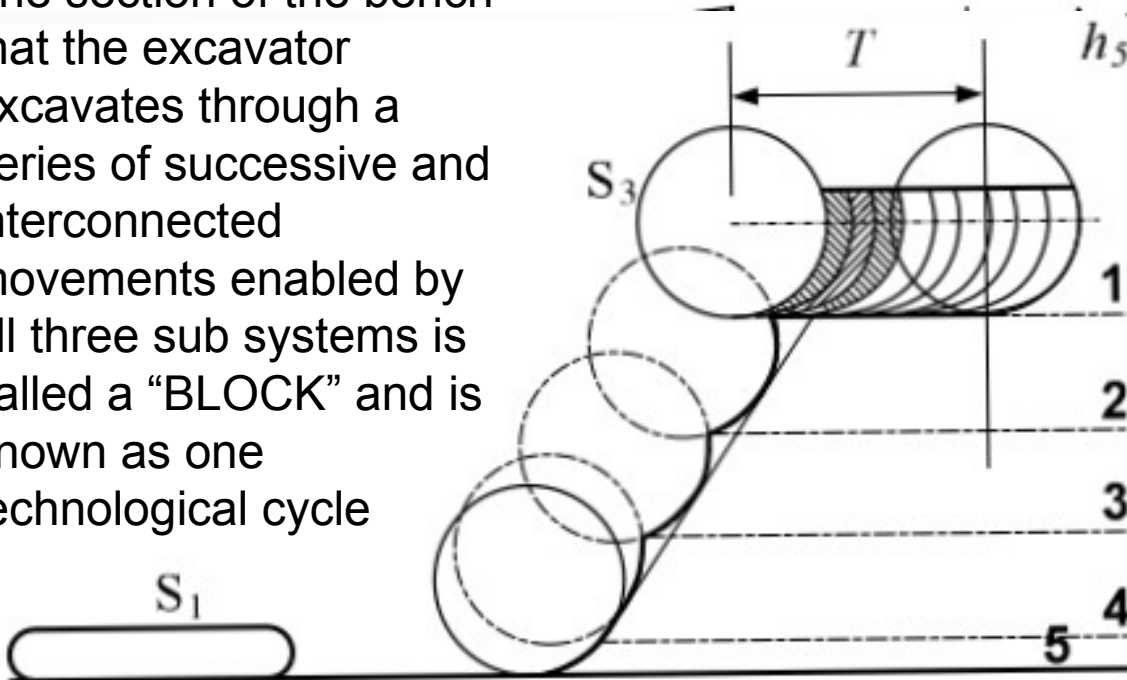
Complimentary equipment and systems



Block / Technological Cycle



The section of the bench that the excavator excavates through a series of successive and interconnected movements enabled by all three sub systems is called a “BLOCK” and is known as one technological cycle



- Subsystem S_1 , transport drive, enables horizontal movement of the excavator along the direct axis.
- Subsystem S_2 , slewing drive, enables rotational movement of the boom together with the upper part of the excavator.
- Subsystem S_3 , bucket wheel drive, enables rotational movement of the bucket wheel around its axis, in the vertical plane.

